Quantum Mechanics 1 - Exam & Quiz Archive

Test 1 - Fall 2025

Useful Expressions provided:

- Time-dependent Schrödinger equation: $i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + V\Psi$
- Time-independent Schrödinger equation: $-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi = E\psi$
- Momentum operator: $\hat{p} = \frac{\hbar}{i}\frac{\partial}{\partial x}$
- Gaussian integral: $\int_{-\infty}^{\infty} e^{-Cx^2}dx = \sqrt{\frac{\pi}{C}}$
- Orthonormality relation: $\int \psi_m^*(x)\psi_n(x)dx = \delta_{mn}$
- Euler's formula: $e^{i\theta} = \cos(\theta) + i\sin(\theta)$
- Double-angle identity for sine squared: $\sin^2(x) = \frac{1-\cos(2x)}{2}$
- Cosine angle-sum identity: $\cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$
- Cosine angle-difference identity: $\cos(x-y) = \cos(x)\cos(y) + \sin(x)\sin(y)$

QUESTION 1 (20 points) A particle of mass m has the wave function:

$$\Psi(x,t) = Ae^{-a[(mx^2/\hbar)+it]}$$

a) (10 points) Find A.

b) (10 points) What is the equation that you need to use to compute the expectation value of x? What is the answer?

QUESTION 2 (10 points) At time $t = 0$, a particle is represented by the wave function:

$$\Psi(x,0) = \begin{cases} A(x/a), & \text{if } 0 \le x \le a \\ A(b-x)/(b-a), & \text{if } a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

where $A = \sqrt{\frac{3}{b}}$, and a and b are (positive) constants. What is the probability of finding the particle to the left of a?

QUESTION 3 (10 points) Why is the following wave function a stationary state?

$$\Psi_1(x,t) = \psi_1(x)e^{-iE_1t/\hbar}$$

Explain your answer using the probability density of $\Psi_1(x,t)$.

QUESTION 4 (20 points) Given the general solution of the time-dependent Schrödinger equation,

$$\Psi(x,t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar}$$

a) (10 points) What is the physical meaning of $|c_n|^2$?

b) (10 points) Write the expectation value of the energy, $\langle H \rangle$, using $|c_n|^2$.

QUESTION 4 (40 points) [Duplicate Numbering in Original] The infinite square well potential is given by:

$$V(x) = \begin{cases} 0, & \text{if } 0 \le x \le a \\ \infty, & \text{otherwise} \end{cases}$$

a) (10 points) Using $k^2 = 2mE/\hbar^2$, write down the time independent Schrödinger equation for this potential and its general solution.

b) (10 points) Use the boundary conditions to find the allowed energies.

c) (10 points) Find the normalized solution of the time independent Schrödinger equation.

d) (10 points) Show that the solutions $\psi_n(x)$ and $\psi_m(x)$ are orthogonal for $n \neq m$.

EXTRA POINTS (This question is OPTIONAL): (10 points) Suppose that a particle in the infinite square well has as its initial wave function an even mixture of the first two stationary states:

$$\Psi(x,0) = \frac{1}{\sqrt{2}}[\psi_1(x) + \psi_2(x)]$$

Find the expectation value of the particle's energy.